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Method for preparing induced pluripotent stem cells by reprogramming somatic cells

Abstract

The present invention provides a method for preparing induced pluripotent stem cells through somatic cell reprogramming and induced pluripotent stem cells obtained therefrom. The present method comprises introducing the factors Oct4 and Nanog as reprogramming-inducing factors into somatic cells to perform reprogramming; followed by culturing the partially or fully reprogrammed somatic cells in a medium comprising specific chemical inducing agents to obtain induced pluripotent stem cells. In the present invention, the combination of different forms of reprogramming-inducing factors and three small-molecule compounds as chemical inducing agents can significantly improve the reprogramming efficiency of human somatic cells and reduce the tumorigenicity of the obtained induced pluripotent stem cells.

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Background/Summary

(1) This application is a national stage filing under 35 U.S.C. § 371 of International Application No. PCT/CN2020/106327, filed on Jul. 31, 2020, which claims priority of Chinese Application No.

201910707125.0, filed on Aug. 1, 2019. The contents of each of these applications are incorporated herein by reference.

TECHNICAL FIELD

(2) The present disclosure belongs to the technical field of stem cells, and particularly relates to a method of preparing induced pluripotent stem cells through somatic cell reprogramming, as well as the induced pluripotent stem cells obtained therefrom.

BACKGROUND

(3) In 2006, Yamanaka's team developed a cocktail consisting of four genes, namely Oct4, Sox2, Klf4 and c-Myc, which was used to successfully reprogram terminally differentiated skin fibroblasts into induced pluripotent stem cells (iPSCs) by viral infection. One year later, James Thomson successfully reprogrammed human fibroblasts into iPSCs by episomal plasmid transfection using a different combination of four factors (Oct4, Sox2, Nanog and Lin28). The above methods, which are relatively simple and stable in technology, break through the ethical limitations of using human embryonic stem cells in medicine, capable of solving the problem of immune rejection in cell transplantation therapy, and greatly expand the application potential of stem cell technology in clinical medicine. In addition, iPSCs technology and functional cells obtained by subsequent differentiation have great potential value in areas such as cell replacement therapy, pathogenesis research, and new drug screening.

(4) However, with the development of iPSCs reprogramming technology, the drawbacks and problems of iPSCs reprogramming by exogenous introduction of transcription factors have gradually become prominent, such as incomplete reprogramming of some cells, low reprogramming efficiency, and oncogenicity of the introduced proto-oncogenes. Therefore, researchers are trying to find alternatives to the four factors and to optimize the reprogramming system, such as replacing oncogenic transcription factors, adding transcription factors or compounds that enhance reprogramming efficiency, or even completely giving up transcription factors and switching to small molecule compounds to induce somatic cell reprogramming, so as to avoid the potential defects to a certain extent.

(5) At present, a variety of small molecule compounds have been found to induce the generation of pluripotent stem cells. Some of these small molecules directly improve the reprogramming efficiency of recipient cells by inhibiting genomic methylation; some of them affect specific signaling pathways, rendering the intermediate transitional cells and partially reprogrammed cells generated during the reprogramming process to be transformed into stably and completely induced pluripotent stem cells. For example, TGF β can induce the expression of EMT-related gene SNAIL in the early stage of reprogramming, thereby inducing EMT and hindering the reprogramming process; on the contrary, TGF β receptor inhibitors can improve the reprogramming efficiency and can be used to replace Sox2 and c-Myc in Yamanaka four factors; cyclic AMP (cAMP) agonists can increase Oct4 expression; glycogen synthase kinase (GSK) inhibitors can increase Nanog expression, thereby improving the reprogramming efficiency of specific cells. Some methods even use a cocktail of purely chemical small molecules without involving any exogenously introduced transcription factors. However, such methods have relatively low reprogramming efficiency and are time-consuming, which cannot meet clinical needs. Some of the methods have not been used for the reprogramming of human cells.

(6) To date, researchers have tried reprogramming techniques on many different types of cells and have achieved success. iPSCs can be obtained by reprogramming using genes, RNAs or proteins of the four factors.

(7) Collectively the characteristics of the above methods, it is important and meaningful to provide a preparation method of adding a combination of reprogramming-inducing factors in different types of vectors and small molecule compounds, so as to improve the reprogramming efficiency, and to obtain iPSCs by safely and effectively reprogramming human somatic cells.

SUMMARY OF THE INVENTION

(8) One purpose of the present invention is to provide a method of preparing induced pluripotent stem cells (iPSCs) by reprogramming of somatic cells.

(9) Another purpose of the present invention is to provide induced pluripotent stem cells (iPSCs) prepared by the method as above.

(10) The present inventors found that by introducing only two transcription factors as reprogramming-inducing factors, accompanied with three small molecule compounds as chemical inducing agents, a simpler and more efficient reprogramming method with broader application potentials as compared to the prior art can be achieved on somatic cells from different origins of the human body, and iPSCs were successfully obtained therefrom, thus completing the present invention.

(11) Accordingly, in a first aspect, the present invention provides a method for preparing induced pluripotent stem cells through somatic cell reprogramming, comprising the following steps: (1) introducing Oct4 and Nanog as reprogramming-inducing factors into somatic cells to perform reprogramming; (2) culturing partially or completely reprogrammed somatic cells obtained in step (1) in the presence of chemical inducing agents to obtain induced pluripotent stem cells (iPSCs), wherein the chemical inducing agents comprise a TGF β receptor inhibitor, a cyclic AMP (cAMP) agonist and a glycogen synthase kinase (GSK) inhibitor.

(12) Preferably, the TGF β receptor inhibitor is 616452, and/or the cyclic AMP (cAMP) agonist is Forskolin, and/or the glycogen synthase kinase (GSK) inhibitor is TD114-2.

(13) In the method of the present invention, preferably, Oct4 and Nanog as the reprogramming-inducing factors can be introduced into somatic cells in the form of their DNAs, in the form of their RNAs, or in the form of their protein products. In the method of the present invention, preferably, the somatic cells can be any cells known in the art, for example, skin-derived cells, blood-derived cells, and can also include urine-derived cells, liver cells, epithelial cells, gastric cells, keratinocytes and the like. In a more preferred embodiment, the skin-derived cells may be skin fibroblasts. In another more preferred embodiment, the blood-derived cells may be erythroid progenitor cells.

(14) In the method of the present invention, preferably, the somatic cells are derived from human, that is, human somatic cells.

(15) In the method of the present invention, preferably, the working concentration of the TGF β receptor inhibitor such as 616452 is 0.1-20 μ M, more preferably 5-10 μ M, and still more preferably 5 μ M.

(16) In the method of the present invention, preferably, the working concentration of the cyclic AMP (cAMP) agonist such as Forskolin is 0.1-50 μ M, more preferably 2-20 μ M, and still more preferably 10 μ M.

(17) In the method of the present invention, preferably, the working concentration of the glycogen synthase kinase (GSK) inhibitor such as TD114-2 is 0.1-20 μ M, more preferably 2-10 μ M, still more preferably 5 μ M.

(18) In a specific embodiment, in step (1), reprogramming is conducted by transfecting the vector(s) comprising the coding sequences of the reprogramming-inducing factors Oct4 and Nanog, respectively, into somatic cells by electroporation or chemical transfection. For example, the transfection can be performed by electroporation. Specifically, the electroporation method may include adding reprogramming vector(s) comprising the inducing factors Oct4 and Nanog to cell suspension, transferring into electroporation cuvette after mixing, placing the electroporation cuvette in the electroporation apparatus, and performing electroporation.

(19) In another specific embodiment, in step (1), reprogramming is conducted by infecting the somatic cells with a virus comprising Oct4 and Nanog reprogramming-inducing factors. Preferably, the virus is Sendai Virus.

(20) In the method of the present invention, preferably, the aforesaid specific chemical inducing agents are added to the culture medium from Day 2 after the introduction of the reprogramming-

inducing factors.

(21) In a second aspect, the present invention provides an induced pluripotent stem cell which is obtained by the method of the first aspect.

(22) In a third aspect, the present invention relates to a combination of exogenously introduced two inducing factors and three chemical inducing agents, for use in reprogramming somatic cells to prepare iPSCs, wherein the exogenously introduced two inducing factors are Oct4 and Nanog, and the three chemical inducing agents consist of a TGF β receptor inhibitor such as 616452, a cyclic AMP (cAMP) agonist such as Forskolin, and a glycogen synthase kinase (GSK) inhibitor such as TD114-2. Specifically, the Oct4 and Nanog can be introduced into somatic cells in the form of their DNAs, RNAs or proteins.

(23) In the present invention, different forms of reprogramming-inducing factors in combination with the three small molecule compounds as chemical inducing agents can significantly improve the reprogramming efficiency of human somatic cells and reduce the tumorigenicity of the obtained iPSCs.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1A shows two brightfield images of iPSCs obtained by the methods as described in Example 1, in which the upper panel of FIG. 1A shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 1A shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture. FIG. 1B shows two brightfield images of iPSCs obtained by the methods as described in Example 2, in which the upper panel of FIG. 1B shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 1B shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture. FIG. 1C shows two brightfield images of iPSCs obtained by the methods as described in Example 3, in which the upper panel of FIG. 1C shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 1C shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture.

(2) FIGS. 2A-E are the detection results of flow cytometry, showing the expression of pluripotent cell markers by iPSCs obtained by reprogramming of skin fibroblasts. (A) SSEA4; (B) Tra-1-81; (C) Tra-1-60; (D) Oct4; (E) Nanog.

(3) FIGS. 3A-B show the results of (A) immunofluorescence staining and (B) qPCR identification of differentiation of three germ layers from iPSCs obtained by reprogramming skin fibroblasts.

(4) FIG. 4A shows two brightfield images of iPSCs obtained by the methods as described in Example 5, in which the upper panel of FIG. 4A shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 4A shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture. FIG. 4B shows two brightfield images of iPSCs obtained by the methods as described in Example 6, in which the upper panel of FIG. 4B shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 4B shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture. FIG. 4C shows two brightfield images of iPSCs obtained by the methods as described in Example 7, in which the upper panel of FIG. 4C shows the iPSC colonies after reprogramming for about 12 days, and the lower panel of FIG. 4C shows the iPSC colonies after picking single colonies and inoculating into a feeder-free system for culture.

(5) FIGS. 5A-E are the detection results of flow cytometry, showing the expression of pluripotent cell markers by iPSCs obtained by reprogramming of erythroid progenitor cells. (A) SSEA4; (B) Tra-1-81; (C) Tra-1-60; (D) Oct4; (E) Nanog.

(6) FIGS. 6A-B show the results of (A) immunofluorescence staining and (B) qPCR identification

of differentiation of three germ layers from iPSCs obtained by reprogramming erythroid progenitor cells.

DETAILED DESCRIPTION

(7) In the method of the present invention, only Oct4 and Nanog are used as reprogramming-inducing factors, and reprogramming is performed by introducing them into somatic cells. Both Oct4 and Nanog are transcription factors that play important roles in maintaining pluripotency. A variety of transcription factors have been identified in the prior art that can be used to induce the reprogramming of somatic cells into induced pluripotent stem cells, such as Oct4, c-Myc, Sox2, and Klf4 consisting the Yamanaka four-factor combination as described above, and Oct4, Sox2, Nanog and Lin28 consisting the James Thomson four-factor combination. But the method of the present invention uses only Oct4 in the Yamanaka four factors and Nanog in the James Thomson four factors. In other words, the method of the present invention does not use any transcription factors other than Oct4 and Nanog as reprogramming-inducing factors.

(8) Oct4 and Nanog can be introduced by a method known in the art for introducing transcription factors. Such methods include, but are not limited to, infecting somatic cells by introducing virus comprising recombinant DNA vector(s), mRNAs or RNAs of the nucleotides encoding Oct4 and Nanog so as to allow the expression of Oct4 and Nanog inducing factors, or by directly introducing Oct4 and Nanog in the form of proteins into somatic cells.

(9) In one embodiment of the method of the present invention, the two reprogramming-inducing factors are introduced into somatic cells in the form of DNA. Specifically, a nucleotide sequence encoding Oct4 and a nucleotide sequence encoding Nanog can be introduced into somatic cells. In a specific embodiment, the nucleotide sequence encoding Oct4 comprises the nucleotide sequence as shown in SEQ ID NO: 6 or is consisted of the nucleotide sequence as shown in SEQ ID NO: 6; or comprises a nucleotide sequence that has at least 90% homology to the nucleotide sequence of SEQ ID NO: 6 and encodes Oct4; or is consisted of a nucleotide sequence that has at least 90% homology to the nucleotide sequence of SEQ ID NO: 6 and encodes Oct4. In a specific embodiment, the nucleotide sequence encoding Nanog comprises the nucleotide sequence as shown in SEQ ID NO: 15 or is consisted of the nucleotide sequence as shown in SEQ ID NO: 15; or comprises a nucleotide sequence that has at least 90% homology to the nucleotide sequence of SEQ ID NO: 15 and encodes Nanog; or is consisted of a nucleotide sequence that has at least 90% homology to the nucleotide sequence of SEQ ID NO: 15 and encodes Nanog.

(10) The nucleotide sequence encoding Oct4 and the nucleotide sequence encoding Nanog can be placed in the same vector or in different vectors. When placed in the same vector, the nucleotide sequence encoding Oct4 and the nucleotide sequence encoding Nanog may be under the control of the same or different regulatory sequences. Regulatory sequences can be selected according to the type of target cell. In a specific embodiment, the nucleotide sequence encoding Oct4 and the nucleotide sequence encoding Nanog are placed in the same vector, such as pcDNA3.1. Additional elements, such as the coding sequence of EBNA1 and OriP sequence, may be comprised in the recombinant vector to increase the efficiency of plasmid replication in cells. In a specific embodiment, the sequence of the recombinant vector used in the present invention is shown in SEQ ID NO: 13 or SEQ ID NO: 24.

(11) Methods for delivering vector(s) comprising nucleotides of interest into somatic cells are known in the art and include, but are not limited to, electroporation, gene gun, lipofection, calcium-mediated transfection. In a specific embodiment, electroporation is used.

(12) In the method of the present invention, the above-mentioned inducing factors Oct4 and Nanog can be expressed in vitro to obtain corresponding proteins, which are then introduced into differentiated cells, thereby achieving the object of the present invention. Techniques for introducing proteins into cells are well known in the art and include, but are not limited to, Tat-delivery and related techniques, electroporation (nucleofection), protein and cellular ligand binding.

(13) Those of ordinary skill in the art can also understand that the DNA sequences of the above-mentioned Oct4 and Nanog inducing factors can be transcribed in vitro, and the obtained mRNAs are directly introduced into differentiated cells to express the corresponding proteins in the cells, thus achieving the purpose of the present invention. Reprogramming can also be performed by infecting somatic cells with Sendai Virus comprising RNAs of the two inducing factors. Sendai Virus is a non-integrating virus that will not integrate into the genome of infected cells and thus has relatively high safety.

(14) After introduction of the reprogramming-inducing factors Oct4 and Nanog into somatic cells, cells were cultured in the presence of one or more of a TGF β receptor inhibitor, a cyclic AMP (cAMP) agonist, and a glycogen synthase kinase (GSK) inhibitor as chemical inducing agent(s) to generate iPSCs. In preferred embodiments, a TGF β receptor inhibitor, a cyclic AMP (cAMP) agonist, and a glycogen synthase kinase (GSK) inhibitor are used.

(15) In a preferred embodiment, the TGF β receptor inhibitor is 616452, whose chemical name is 2-[3-(6-methyl-2-pyridinyl)-1H-pyrazol-4-yl]-1,5-naphthyridine (CAS No: 446859-33-2). Preferably, the TGF β receptor inhibitor is used at a working concentration of 0.1-20 μ M, more preferably 5-10 μ M, still more preferably 5 μ M. For example, the TGF β receptor inhibitor is added to the medium at such concentration.

(16) In a preferred embodiment, the cyclic AMP (cAMP) agonist is Forskolin, whose chemical name is [(3R,4aR,5S,6S,6aS,10S,10aR,10bS)-3-ethenyl-6,10,10b-trihydroxy-3,4a,7,7,10a-pentamethyl-1-oxo-5,6,6a,8,9,10-hexahydro-2H-benzo[f]chromen-5-yl]acetate (CAS No: 66575-29-9). Preferably, the cAMP agonist is used at a working concentration of 0.1-50 μ M, more preferably 2-20 μ M, still more preferably 10 μ M. For example, the cAMP agonist is added to the medium at such concentration.

(17) In a preferred embodiment, the glycogen synthase kinase (GSK) inhibitor is TD114-2, whose chemical name is 6,7,9,10,12,13,15,16,18,19-Decahydro-5,29:20,25-dimetheno-26H-dibenzo[n,t]pyrrolo[3,4-q][1,4,7,10,13,22] tetraoxa diazacyclotetracosine-26,28(27H)-dione (CAS No: 436866-52-3). Preferably, the GSK inhibitor is used at a working concentration of 0.1-20 μ M, more preferably 2-10 μ M, still more preferably 5 μ M. For example, the GSK inhibitor is added to the medium at such concentration.

(18) The medium used in the method of the present invention can be selected by a person skilled in the art based on existing knowledge depending on the type of cells to be cultured.

EXAMPLES

(19) In order to facilitate understanding of the present invention, the technical solutions of the present invention will be further exemplified below through specific examples. Those of ordinary skill in the art can understand that the present invention is not limited to the described examples, and those of ordinary skill in the art can make modifications to the examples based on the teachings of the disclosure. Such modifications are also included within the scope of the present invention as defined by the appended claims.

(20) The experimental methods in the following examples are conventional methods unless otherwise specified.

Example 1. Inducing the Reprogramming of Skin Cells by Using a Vector Encoding Reprogramming Factors

(21) 1.1 Human-derived skin tissue was placed into a petri dish, and was quickly and repeatedly rinsed for 4 times with phosphate buffered saline (PBS) pre-cooled at 4° C.

(22) 1.2 The skin tissue was processed with sterilized ophthalmic scissors and a scalpel to remove the subcutaneous white fat, remove the epidermis and subcutaneous tissue, and leave the dermis layer to obtain pretreated skin tissues. The pretreated skin tissue was transferred in the medium, and cut into small pieces to obtain skin tissue pieces with neat edges.

(23) 1.3 3 mL of fetal bovine serum (FBS) was evenly added to each well of a 6-well cell culture plate. The skin tissue pieces were added and incubated in the incubator for 0.5-1 h to make the

tissues adhere to the bottom of the culture plate.

(24) 1.4 1 mL of DMEM (Gibco) medium comprising 20% (v/v) FBS was added to each well of the 6-well cell culture plate, and the plate was put back into the incubator for culture.

(25) 1.5 The medium was exchanged when fibroblasts migrated out of the skin tissue pieces as observed under the microscope. The amount of DMEM medium was increased to 3 mL, and the medium was exchanged every 1-3 days.

(26) 1.6 When the fibroblasts converged to the edge of each well, the spent medium was discarded. After washing twice with PBS, 1 mL of 0.25% trypsin-EDTA (GIBCO) was added, and the plate was placed in a cell culture incubator for about 4-6 min.

(27) When the cells became round, detached and floated, DMEM medium comprising 20% (v/v) FBS was immediately added to terminate the digestion. The cells were pipetted several times gently and transferred into a 15 mL centrifuge tube, and centrifuged at 200 g for 3 min. The supernatant was aspirated and discarded. The cell culture medium was added to the tube and mixed thoroughly. The cells were transferred to a culture flask, which was placed in a cell incubator to continue culturing.

(28) 1.7 When the second-generation skin fibroblasts reached 80%-90% confluence, the old medium was discarded. The cells were washed twice with PBS, followed by digestion via adding 3 mL of 0.25% trypsin-EDTA (Gibco), until the cells were dispersed. DMEM medium comprising 20% (v/v) FBS was added to stop digestion. The cells were pipetted several times gently and transferred into a 15 mL centrifuge tube, and centrifuged at 200 g for 3 min. The supernatant was discarded. A suitable amount of PBS was added to resuspend the cells, which were counted by using a hemocytometer.

(29) 1.8 About 8×10^5 skin fibroblasts were taken for centrifuge at 200 g for 3 min. The cells were washed twice with PBS and once with OPTI-MEM (Gibco). The supernatant was discarded. Corresponding electroporation reagents were added according to the instructions of the Cellectra Kit. Cells were resuspended. The reprogramming vector comprising the inducing factors Oct4 and Nanog were added to the cell suspension, mixed well and transferred to an electroporation cuvette. The electroporation cuvette was placed in an electroporation apparatus, and electroporation was conducted at 430 V, 30 ms.

(30) Specifically, the reprogramming vector was constructed as follows: {circle around (1)} Using the plasmid (Addgen #20922) as the template, KOD-Plus-Neo (TOYOBO #KOD-401) high-fidelity enzyme and primers F1/R1, F2/R2, F3/R3 were used to perform the amplification, resulting in the fragment of EF-1 α promoter, Oct4 coding sequence and Nanog coding sequence, respectively (see Table 1 below for the primer sequences and the sequences obtained by the amplification). {circle around (2)} Using the pcDNA3.1(-) plasmid as template, amplification was performed by using KOD-Plus-Neo (TOYOBO #KOD-401) high-fidelity enzyme and primers F4/R4 to obtain the pcDNA3.1 fragment (see Table 1 below for the primer sequences and the sequence obtained by the amplification). {circle around (3)} Using NEBuilder HiFi DNA Assembly Bundle for Large Fragments (NEB #E2623), the pcDNA3.1 was ligated with the EF-1 α promoter, the Oct4 coding sequence and the Nanog coding sequence by homologous recombination, so that the coding sequences of Oct4 and Nanog were under the control of the EF-1 α promoter, resulting in the target vector, namely pcDNA3.1-EF-Oct4-Nanog (SEQ ID NO: 13).

(31) 1.9 After the electroporation, the electroporation cuvette was taken out. The cell suspension was quickly aspirated and added to Matrigel (Corning)-coated cell culture plate comprising 2 mL of DMEM medium comprising 20% (v/v) FBS added in advance. On Day 2, the medium was changed to 2-3 mL of TeSR-E7 medium (Stem Cell Technologies). Static culture was conducted at 37° C. in cell incubator. The medium was completely exchanged every day since then. Starting from Day 2, medium being used was supplemented with a combination of small molecule compounds, in which the working concentrations of the compounds were as follows. The working concentration of TGF β receptor inhibitor 616452 was 5 μ M. The working concentration of cyclic

AMP (cAMP) agonist Forskolin was 10 μ M. The working concentration of glycogen synthase kinase (GSK) inhibitor TD114-2 was 5 μ M.

(32) After reprogramming for about 12 days, iPSC colonies with cell morphology distinct from those of skin fibroblasts were observed (see the upper panel of FIG. 1A). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 1A).

Example 2. Inducing the Reprogramming of Skin Cells by Using a Vector Encoding Reprogramming Factors

(33) The same protocol was used as described in steps 1.1-1.9 of Example 1. Fibroblasts derived from human skin tissue were used for conducting the method of the present invention. As compared to Example 1, the construction of the reprogramming vector in step 1.8 is different. On the basis of the construct of Example 1, EBNA1 and OriP elements were added, so that the replication efficiency of the plasmid could be enhanced in somatic cells based on the EBNA1/OriP system. The construction method is described in detail as follows.

(34) {circle around (1)} Using plasmid (Addgen #20922) as the template, KOD-Plus-Neo (TOYOBO #KOD-401) high-fidelity enzyme and primers F5/R5, F6/R6, F7/R7, F8/R8, F9/R9 were used to perform the amplification, resulting in the fragment of EF-1 α promoter, Oct4 coding sequence, Nanog coding sequence, EBNA1 coding sequence and OriP sequence, respectively (see Table 1 below for the primer sequences and the sequences obtained by the amplification). {circle around (2)} Using the pcDNA3.1(-) plasmid as the template, amplification was performed by using KOD-Plus-Neo (TOYOBO #KOD-401) high-fidelity enzyme and primers F10/R10 to obtain the pcDNA3.1 fragment (see Table 2 below for the primer sequences and the sequence obtained by the amplification). {circle around (3)} Using NEBuilder HiFi DNA Assembly Bundle for Large Fragments (NEB #E2623), the pcDNA3.1 was ligated with the EF-1 α promoter, the Oct4 coding sequence, the Nanog coding sequence, the EBNA1 coding sequence and the OriP sequence by homologous recombination, resulting in the target vector, namely pcDNA3.1-EF-Oct4-Nanog-EBNA1-OriP (SEQ ID NO: 24).

(35) After reprogramming for about 12 days, iPSC colonies with cell morphology distinct from those of skin fibroblasts were observed (see the upper panel of FIG. 1B). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 1B).

Example 3. Inducing the Reprogramming of Skin Cells by Using a Virus Comprising RNAs of Reprogramming-Inducing Factors

(36) The same protocol was used as described in steps 1.1-1.6 of Example 1. Fibroblasts derived from human skin tissue were used for conducting the method of the present invention. Different from steps 1.7-1.9 of Example 1, reprogramming of iPSCs was performed by using a virus comprising RNAs of reprogramming-inducing factors as described in step 3.7 below.

(37) 3.7 When the second-generation skin fibroblasts reached 80%-90% confluence, Sendai Virus comprising Oct4 and Nanog reprogramming-inducing factors was mixed with the cells and cultured for 2 days. The medium was discarded. 2 mL of DMEM medium with 20% (v/v) FBS was added to culture, and the medium was changed every other day. After culturing for 5 days, 1 mL of 0.25% trypsin-EDTA (GIBCO) was added, followed by culture in a cell incubator for about 4~6 min. When the cells became round, detached and floated, they were added to Vitronectin (Gibco)-coated cell culture plates for culture. Starting from Day 2 of the culture, medium being used was supplemented with a combination of small molecule compounds, in which the working concentrations of the compounds were as follows. The working concentration of TGF β receptor

inhibitor 616452 was 5 μ M. The working concentration of cyclic AMP (cAMP) agonist Forskolin was 10 μ M. The working concentration of glycogen synthase kinase (GSK) inhibitor TD114-2 was 5 μ M.

(38) After reprogramming for about 12 days, iPSC colonies with cell morphology distinct from those of skin fibroblasts were observed (see the upper panel of FIG. 1C). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 1C).

Example 4. Identification of iPSCs Obtained by Reprogramming Skin Fibroblasts

(39) The iPSCs obtained by reprogramming skin fibroblasts were characterized and identified using different methods, including flow cytometry, immunofluorescence staining and quantitative PCR (qPCR).

(40) The obtained iPSCs were analyzed by flow cytometry for the following molecular markers: SSEA4, Tra-1-81, Tra-1-60, Oct4 and Nanog. As shown in FIG. 2, the iPSCs obtained by the method of the present invention expressed the markers of human pluripotent stem cells including SSEA4, Tra-1-81, Tra-1-60, Oct4 and Nanog, which proved that the obtained cells possessed the characteristics of pluripotent stem cells.

(41) To verify the totipotency, the obtained iPSCs were induced to differentiate into three germ layers in vitro. The expression of molecular markers specific for the three germ layers was detected by immunofluorescence staining and qPCR. As shown in FIG. 3A, results of the immunofluorescence experiments using three markers (endoderm: SOX17; mesoderm: CDX2; ectoderm: PAX6) showed that the obtained iPSCs could differentiate into cells of all three germ layers. FIG. 3B shows the expression of molecular markers specific for three germ layers (endoderm: SOX17, mesoderm: MIXL1, and ectoderm: PAX6) detected by qPCR, in which the iPSCs expressed the molecular markers of pluripotent stem cell, OCT4 and TRA-1-81, at high levels (consistent with the aforementioned results of flow cytometry assay), while cells differentiated into three germ layers expressed the molecular markers specific for each germ layer (**** $p < 0.0001$, Student's t-test). These results indicated that the obtained iPSCs were totipotent.

Example 5. Inducing the Reprogramming of Peripheral Blood Cells by Using a Vector Encoding Reprogramming Factors

(42) 5.1 10 mL of human peripheral blood was collected, from which the erythroid progenitor cells were enriched and expanded by using RosetteSep™ kit and SepMate™ kit (Stem Cell Technologies). Specifically, 10 mL of blood was transferred from the blood collection tube to a regular centrifuge tube, to which 50 μ L of RosetteSep™ Human Progenitor Cell Basic Pre-Enrichment Cocktail was added and mixed thoroughly. Then the tubes were placed at room temperature for 10 min.

(43) 5.2 3.5 mL of Lymphoprep™ was added to a SepMate centrifuge tube along the central hole. After incubating at room temperature for 10 minutes, 10 mL of PBS+2% FBS was added and mixed thoroughly. 5 mL of blood was added along the wall of the SepMate centrifuge tube. Immediately after centrifuge at 1200 g for 10 min, the yellow supernatant was poured carefully into a new regular centrifuge tube (be careful not to pour in the unwanted cells at bottom), and centrifuged at 300 g for 8 min.

(44) 5.3 After centrifugation, the supernatant was discarded. The cells were resuspended with 0.5 mL of StemSpan™ SFEM II medium and counted for the total number. 2 mL of cells at a density of 5×10^6 /mL was inoculated into each well of a 6-well plate, and cultured at 37° C., 5% CO₂ in cell incubator.

(45) 5.4 Day 1: the cell suspension was transferred to a new 6-well plate to remove adherent unwanted cells. Each well was supplemented with 0.5 mL StemSpan™ SFEM II medium, and incubated at 37° C., 5% CO₂ in cell incubator.

(46) 5.5 Day 2, Day 4, Day 6 and Day 8: Cell suspension from each well was collected into a centrifuge tube. Centrifuge was conducted at 400 g for 5 min, and the supernatant was discarded. StemSpan™ SFEM II medium was added to resuspend the cells. After pipetting for 3-4 times, 2 mL was inoculated into each well of a new 6-well plate, shaken thoroughly and cultured at 37° C., 5% CO₂ in cell incubator.

(47) 5.6 Day 9: The number of erythroid progenitor cells significantly increased. Further, the erythroid progenitor cells rapidly enriched on Day 10. When the number of cells reached 2×10^6 , electroporation could be performed. The cell suspension was collected and centrifuged at 400 g for 5 min. The supernatant was discarded. According to the instructions of the Cellectra Kit, the corresponding electroporation reagents were added to resuspend the cells. Into the cell suspension, the reprogramming vector comprising inducing factors Oct4 and Nanog as constructed in Example 1, namely pcDNA3.1-EF-Oct4-Nanog (SEQ ID NO: 13), was added. After mixing thoroughly, the mixture was added to a electroporation cuvette, which was placed in electroporation apparatus to perform the electroporation at 820 V, 20 ms.

(48) 5.7 After the electroporation, the electroporation cuvette was taken out. The cell suspension was quickly aspirated and added to Matrigel (Corning)-coated cell culture plate comprising StemSpan™ SFEM II added in advance. On Day 2, 1 mL StemSpan™ SFEM II medium was supplemented to each well. Static culture was conducted at 37° C. in cell incubator. On Day 3 and Day 5, 1 mL ReproTeSR medium (Stem Cell Technologies) was supplemented to each well. On Day 7, it could be observed that cells began to grow adherent to the wall. The spent medium was discarded. Afterwards, the medium was replaced with 2 mL of fresh ReproTeSR medium for each well every day. Starting from Day 2, medium being used was supplemented with a combination of small molecule compounds, in which the working concentrations of the compounds were as follows. The working concentration of TGFβ receptor inhibitor 616452 was 5 μM. The working concentration of cyclic AMP (cAMP) agonist Forskolin was 10 μM. The working concentration of glycogen synthase kinase (GSK) inhibitor TD114-2 was 5 μM. After reprogramming for about 12 days, iPSC colonies derived from adherent growth of the cell mass in suspension culture could be observed (see the upper panel of FIG. 4A). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 4A). Example 6. Inducing the Reprogramming of Peripheral Blood Cells by Using a Vector Encoding Reprogramming Factors

(49) The same protocol was used as described in steps 5.1-5.7 of Example 5. Erythroid progenitor cells derived from human peripheral blood were used for conducting the method of the present invention. Different from Example 5, the reprogramming vector used in step 5.6 was replaced by the reprogramming vector comprising inducing factors Oct4 and Nanog as constructed in Example 2, namely pcDNA3.1-EF-Oct4-Nanog-EBNA1-OriP.

(50) After reprogramming for about 12 days, iPSC colonies derived from adherent growth of the cell mass in suspension culture could be observed (see the upper panel of FIG. 4B). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 4B).

Example 7. Inducing the Reprogramming of Peripheral Blood Cells by Using a Virus Comprising RNAs of Reprogramming-Inducing Factors

(51) The same protocol was used as described in steps 5.1-5.5 of Example 5. Erythroid progenitor cells derived from human peripheral blood were used for conducting the method of the present invention. Different from steps 5.6-5.7 of Example 5, reprogramming of iPSCs was performed by using a virus comprising RNAs of reprogramming-inducing factors as described in step 7.6 below.

(52) 7.6 On Day 9: The number of erythroid progenitor cells significantly increased. The cell suspension was collected and centrifuged at 400 g for 5 min. The supernatant was discarded. Sendai Virus comprising Oct4 and Nanog reprogramming-inducing factors in the form of RNAs was added to the cells and mixed. The cells were cultured for 2 days. The medium was discarded. 2 mL of ReproTeSR medium was added. Afterwards, the medium was replaced with 2 mL of fresh ReproTeSR medium for each well every day. Starting from Day 2, medium was supplemented with a combination of small molecule compounds, in which the working concentrations of the compounds were as follows. The working concentration of TGF β receptor inhibitor 616452 was 5 μ M. The working concentration of cyclic AMP (cAMP) agonist Forskolin was 10 μ M. The working concentration of glycogen synthase kinase (GSK) inhibitor TD114-2 was 5 μ M. After reprogramming for about 12 days, iPSC colonies derived from adherent growth of the cell mass in suspension culture could be observed (see the upper panel of FIG. 4C). Single colonies were picked and inoculated into a feeder-free system for culture. These iPSC colonies formed by cells with relatively small size were compact and well-defined iPSC colonies, in which the cells had a relatively large nuclei and high nucleocytoplasmic ratio, showing a typical iPSC morphology (see the lower panel of FIG. 4C).

Example 8. Identification of iPSCs Obtained by Reprogramming Erythroid Progenitor Cells

(53) The iPSCs obtained by reprogramming erythroid progenitor cells were characterized and identified using different methods, including flow cytometry, immunofluorescence staining and quantitative PCR (qPCR).

(54) The obtained iPSCs were analyzed by flow cytometry for the following molecular markers: SSEA4, Tra-1-81, Tra-1-60, Oct4 and Nanog. As shown in FIG. 5, the iPSCs obtained by the method of the present invention expressed the markers of human pluripotent stem cells including SSEA4, Tra-1-81, Tra-1-60, Oct4 and Nanog, which proved that the obtained cells possessed the characteristics of pluripotent stem cells.

(55) To verify the totipotency, the obtained iPSCs were induced to differentiate into three germ layers in vitro. The expression of molecular markers specific for the three germ layers was detected by immunofluorescence staining and qPCR. As shown in FIG. 6A, results of the immunofluorescence experiments using three markers (endoderm: SOX17; mesoderm: CDX2; ectoderm: PAX6) showed that the obtained iPSCs could differentiate into cells of all three germ layers. FIG. 6B shows the expression of molecular markers specific for three germ layers (endoderm: SOX17, mesoderm: MIXL1, and ectoderm: PAX6) detected by qPCR, in which the iPSCs expressed the molecular markers of pluripotent stem cell, OCT4 and TRA-1-81, at high levels (consistent with the aforementioned results of flow cytometry assay), while cells differentiated into three germ layers expressed the molecular markers specific for each germ layer (****p<0.0001, Student's t-test). These results indicated that the obtained iPSCs were totipotent.

(56) TABLE-US-00001 TABLE 1 Products of PCR amplification Primer Sequence after No. Gene name name 5'-3' bp amplification 1 Ef-1 α F1 SEQ ID NO: 1 35 SEQ ID NO: 3 R1 SEQ ID NO: 2 35 2 Oct4 F2 SEQ ID NO: 4 20 SEQ ID NO: 6 R2 SEQ ID NO: 5 20 3 Nanog F3 SEQ ID NO: 7 92 SEQ ID NO: 9 R3 SEQ ID NO: 8 35 4 pcDNA3.1 F4 SEQ ID NO: 10 20 SEQ ID NO: 12 R4 SEQ ID NO: 11 20

(57) TABLE-US-00002 TABLE 2 Products of PCR amplification Gene Sequence after No. name Primer name 5'-3' bp amplification 1 EF-1 α F5, identical SEQ ID NO: 1 35 SEQ ID NO: 3 to F1 R5, identical SEQ ID NO: 2 35 to R1 2 Oct4 F6, identical SEQ ID NO: 4 20 SEQ ID NO: 6 to F2 R6, identical SEQ ID NO: 5 20 to R2 3 Nanog F7, identical SEQ ID NO: 7 92 SEQ ID NO: 15 to F3 R7 SEQ ID NO: 14 92 4 EBNA1 F8 SEQ ID NO: 16 38 SEQ ID NO: 18 R8 SEQ ID NO: 17 85 5 OriP F9 SEQ ID NO: 19 101 SEQ ID NO: 21 R9 SEQ ID NO: 20 34 6 pcDNA3.1 F10 SEQ ID NO: 22 20 SEQ ID NO: 23 R10, identical SEQ ID NO: 11 20 to R4

(58) The method for preparing iPSCs of the present invention reduces the number of transcription factors required for reprogramming to two, reduces the tumorigenicity, and improves the

transformation efficiency through the combination of three small molecule compounds. In general, the method of the present invention is simple, efficient and easy to operate. The iPSCs prepared by this method would be more suitable for clinical translations and applications.

(59) The above are only the preferred embodiments of the present invention. It should be pointed out that for those skilled in the art, without departing from the principles of the present invention, several improvements and modifications can be made. These improvements and modifications should also be regarded as within the protection scope of the present invention.

Claims

1. A method of generating mammalian induced pluripotent stem cells (iPSCs), the method comprising: (1) introducing one or more nucleic acids encoding the reprogramming-inducing factors Oct4 and Nanog into mammalian fibroblasts or mammalian erythroid progenitor cells to obtain reprogrammed fibroblasts or erythroid progenitor cells; (2) culturing the reprogrammed fibroblasts or erythroid progenitor cells obtained in step (1) in the presence of chemical inducing agents to obtain induced pluripotent stem cells (iPSCs), wherein the chemical inducing agents are consisting of a TGF β receptor inhibitor, a cyclic AMP (cAMP) agonist, and a glycogen synthase kinase (GSK) inhibitor, wherein only Oct4 and Nanog are used as reprogramming-inducing factors.
2. The method according to claim 1, wherein the reprogramming-inducing factors Oct4 and Nanog are introduced in the mammalian fibroblasts or mammalian erythroid progenitor cells in the form of DNA thereof or in the form of mRNA thereof.
3. The method according to claim 1, wherein the TGF β receptor inhibitor is 616452, and/or the cyclic AMP (CAMP) agonist is Forskolin, and/or the glycogen synthase kinase (GSK) inhibitor is TD114-2.
4. The method according to claim 3, wherein the working concentration of the TGF β receptor inhibitor 616452 is 0.1-20 μ M; and/or the working concentration of the cyclic AMP (CAMP) agonist Forskolin is 0.1-50 μ M; and/or the working concentration of the glycogen synthase kinase (GSK) inhibitor TD114-2 is 0.1-20 μ M.
5. The method according to claim 1, wherein the mammalian fibroblast or mammalian erythroid progenitor cell is isolated from skin cells, blood cells, urine cells, liver cells, epithelial cells, or gastric cells.
6. The method according to claim 1, wherein the mammalian fibroblasts or mammalian erythroid progenitor cells are isolated from human.
7. The method according to claim 1, wherein in step (1), the introducing one or more nucleic acids encoding the reprogramming-inducing factors Oct4 and Nanog into the mammalian fibroblasts or mammalian erythroid progenitor cells comprises introducing: a) a single recombinant vector comprising both nucleotide sequences encoding the reprogramming-inducing factors Oct4 and Nanog, or b) two recombinant vectors, wherein one recombinant vector comprises a nucleic acid sequence encoding Oct4 and the other recombinant vector comprises a nucleic acid sequence encoding Nanog.
8. The method according to claim 1, wherein in step (1), the introducing one or more nucleic acids encoding the reprogramming-inducing factors Oct4 and Nanog is performed by infecting the mammalian fibroblasts or mammalian erythroid progenitor cells with: a) a single recombinant Sendai virus comprising a nucleotide sequence encoding Oct4 and a nucleotide sequence encoding Nanog, or b) two recombinant Sendai viruses, wherein one virus comprises the nucleotide sequence encoding Oct4 and the other virus comprises the nucleotide sequence encoding Nanog.
9. The method according to claim 8, wherein at least one of the nucleotide sequence encoding Oct4 and the nucleotide sequence encoding Nanog is RNA.
10. The method according to claim 4, wherein the working concentration of 616452 is 5-10 μ M.
11. The method according to claim 4, wherein the working concentration of 616452 is 5 μ M.

12. The method according to claim 4, wherein the working concentration of Forskolin is 2-20 μM .
 13. The method according to claim 4, wherein the working concentration of Forskolin is 10 μM .
 14. The method according to claim 4, wherein the working concentration of TD114-2 is 2-10 μM .
 15. The method according to claim 4, wherein the working concentration of TD114-2 is 5 μM .
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